Chapter 1. First-order Linear Differential Equations
Sections 1.2, 1.4, 1.9, 1.10

1. Consider the first-order linear homgeneous differential equation:

$$\frac{dy}{dt} + a(t)y = 0. \tag{1}$$

The general solution of (1) is given by

$$y(t) = ce^{-\int a(t)\,dt}.$$

The specific solution of the initial-value problem

$$\frac{dy}{dt} + a(t)y = 0; \quad y(t_0) = y_0 \tag{2}$$

is given by

$$y(t) = y_0 e^{-\int_{t_0}^{t} a(s)\,ds}.$$

2. Consider the first-order linear nonhomgeneous differential equation:

$$\frac{dy}{dt} + a(t)y = b(t). \tag{3}$$

Using the integrating factor $\mu(t)y = e^{\int a(t)\,dt}$, we get

$$\frac{d}{dt}\mu(t)y = \mu(t)b(t)$$

Then, the general solution of (3) is given by

$$\begin{aligned} y(t) &= \frac{1}{\mu(t)}\left(\int \mu(t)b(t)\,dt + c\right) \\ &= e^{-\int a(t)\,dt}\left(\int \mu(t)b(t)\,dt + c\right) \end{aligned}$$

The specific solution of the initial-value problem

$$\frac{dy}{dt} + a(t)y = b(t); \quad y(t_0) = y_0 \tag{4}$$

is given by

$$y(t) = \frac{1}{\mu(t)}\left(u(t_0)y_0 + \int_{t_0}^{t} \mu(s)b(s)\,ds\right).$$

3. Consider the first-order general differentail equation of seperable form:

$$\frac{dy}{dt} = \frac{g(t)}{f(y)}. \tag{5}$$

The general solution of (5) is given by

$$F(y(t)) = \int g(t)\ dt + c \tag{6}$$

where $F(y) = \int f(y)\ dy$. We *may* then solve $y = y(t)$ from (6). Now consider the corresponding initial-value problem:

$$\frac{dy}{dt} = \frac{g(t)}{f(y)}; \quad y(t_0) = y_0. \tag{7}$$

It turns out that

$$\int_{y_0}^{y(t)} f(r)\ dr = \int_{t_0}^{t} g(s)\ ds, \tag{8}$$

We *may* then solve the specific solution $y = y(t)$ from (8). This equation is nonlinear, so that we also need to determine the interval of existence of the solution.

4. Consider the first-order differential equation of the form

$$M(t, y) + N(t, y)\frac{dy}{dt} = 0. \tag{9}$$

- The equation (9) is *exact* if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial t}. \tag{10}$$

- If (9) is exact then, it can be transformed into the equation

$$\frac{d\phi}{dt} = 0$$

where

$$\phi(t, y) = \int M(t, y)\ dt + \int \left[N(t, y) - \int \frac{\partial M(t, y)}{\partial y}\ dt\right]\ dy$$

or equivalently

$$\phi(t, y) = \int N(t, y)\ dy + \int \left[M(t, y) - \int \frac{\partial N(t, y)}{\partial t}\ dy\right]\ dt.$$

Consequently, the general solution of (9) is given by

$$\phi(t) = c.$$

There are three common methods to find ϕ, see Pages 60-61.

5. If the equation (9) is not exact, but

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial t}}{N} = R(t)$$

is a function of t alone, then using the integrating factor $\mu(t) = e^{\int R(t)\, dt}$, we can get an exact equation

$$\mu(t)M(t,y) + \mu(t)N(t,y)\frac{dy}{dt} = 0$$

that have the same solution as (9).

6. Consider the initial-value problem

$$\frac{dy}{dt} = f(t,y); \quad y(t_0) = y_0. \tag{11}$$

- It holds that

$$y(t) = y_0 + \int_{t_0}^{t} f(s, y(s))\, ds. \tag{12}$$

  Based on the equation (12), the Picard's iterates of (11) is defined to be:

$$y_{n+1}(t) = y_0 + \int_{t_0}^{t} f(s, y_n(s))\, ds, \quad n = 0, 1, 2, \cdots$$

  with $y_0(t) = y_0$.

- Let the rectangle $R := [t_0, t_0 + a] \times [y_0 - b, y_0 + b]$ and set

$$M = \max_{(r,y)\in R} |f(t,y)|, \quad \alpha = \min(a, \frac{b}{M}).$$

  Then the problem (11) has at least one solution $y(t)$ on the interval $[t_0, t_0 + \alpha]$.

- You must know how to choose suitable a and b to obtain the maximal α, see Pages 79-80.